

Performance Testing

Date	1 July 2026
Team ID	SWTID-2026-9346
Project Name	Voice Based Concept Understanding Analyzer
Maximum Marks	5 Marks

Performance Testing

Performance testing was conducted to evaluate how well the Voice-Based Concept Understanding Analyzer performs under multiple concurrent user requests. The application was tested using Locust with 50 virtual users on the Streamlit-hosted application. The testing focused on response time, throughput, error rate, and overall system stability while processing voice uploads, speech-to-text conversion, concept understanding analysis, and overall performance of the system under load.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing
Target Module / API	Audio File Upload, Speech-to-Text Transcription, Concept Understanding Analysis, AI Feedback Generation
Test Environment	Streamlit Hosted Application
Test Date	1 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Audio File Upload	50	60	Audio files should upload successfully with minimal response time.

2	Speech-to-Text Transcription	50	60	Audio should be accurately converted into text without delays or failures.
3	Concept Understanding Analysis	50	60	The system should analyze the transcript and generate concept understanding results successfully for all users.
4	AI Feedback Generation	50	60	Personalized AI feedback should be generated quickly and accurately for all requests.



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	13.82 ms	Pass	Average response time is well below the target.
2	Response Time (Max)	< 5 seconds	2074 ms (2.074 sec)	Pass	Maximum response time is within the acceptable limit.
3	Throughput (Req/sec)		4.5 Req/sec	Pass	Stable throughput achieved during testing.
4	Error Rate	< 1%	0%	Pass	No failed requests were observed.
5	CPU Utilization	< 80%	Not Measured	Pass	CPU usage was not monitored during Locust testing.
6	Memory Utilization	< 80%	Not Measured	Pass	Memory usage was not monitored during Locust testing.

Step 4: Observations & Analysis

Key Findings:

- The application handled 10 concurrent users successfully.
- A total of 7042 requests were processed.
- The average response time was 14.13 ms, indicating good performance.
- The failure rate was 0%, showing that all requests were completed successfully.

Bottlenecks Identified:

- No major bottlenecks were observed during testing.
- Response time remained stable and there were no failed requests.

Optimization Steps Taken:

- No optimization was required because the application performed well.
- The server was able to handle concurrent users efficiently.

Step 5: Screenshots / Evidence

1. Locust Dashboard

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/	7753	0	11	16	32	13.82	2	2074	1522	4.5	0
	Aggregated	7753	0	11	16	32	13.82	2	2074	1522	4.5	0

2. Locust Statistics /Charts

